
Subnetting

Problem

Assume that a large number of consecutive IP addresses are available starting at 198.16.32.0 and suppose that four organizations, A, B, C, and D, request 1000, 3000, 2000, and 4500 addresses, respectively, in that order.

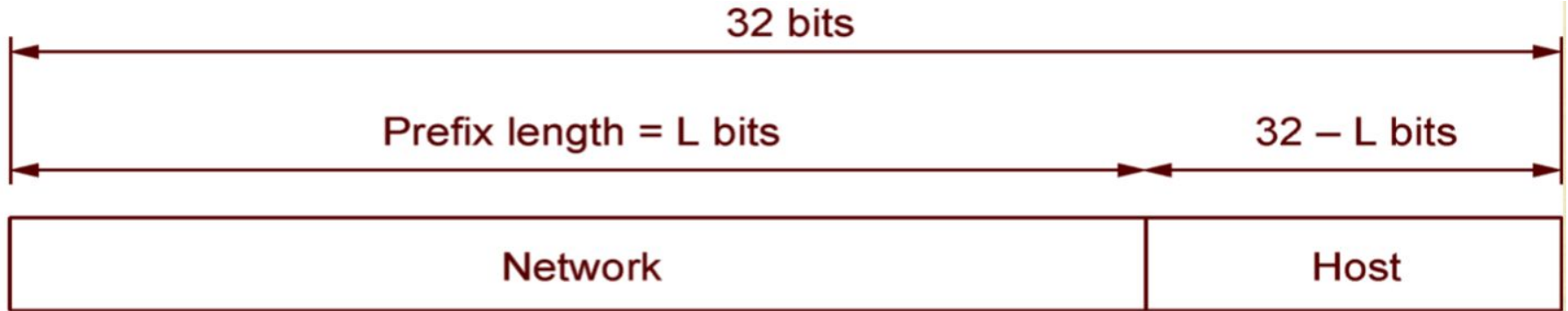
For each of these, give the first possible IP address within the assigned subnet, the last possible IP address within the assigned subnet, and the subnet representation in slash notation (w.x.y.z/s).

Breakdown

1. Initial IP:
 - a. 198.16.32.0
2. Divide into:
 - a. 1000
 - b. 3000
 - c. 2000
 - d. 4500

Subnetting refresher

- Assign range of IP addresses to organization.
- Partition blocks of addresses based on starting IP address given.



Basic steps of subnetting

1. Get the subnet mask (slash notation)
 - a. Power of 2
2. Verify starting address.
 - a. “n” number of bits needed
3. Get last address

Results

Organization	First possible IP Addr. in Assigned subnet	Last possible IP Addr. in Assigned subnet	Subnet representation (Slash Notation)
Business A			
Business B			
Business C			
Business D			

Business A

Starting IP Address

198.16.32.0

Find the nearest power of 2

1000 needed

Powers of 2 Table

Bit Line #	Powers of 2 Exponent	Binary Bit Weight in Decimal	Highest Number Count (Memory Address)		
			Decimal	Hexadecimal	Binary
1	2^0	1	1	0001	0000 0001
2	2^1	2	3	0003	0000 0011
3	2^2	4	7	0007	0000 0111
4	2^3	8	15	000F	0000 1111
...	...	...	...	...	...
9	2^8	256	511	0000 01FF	0001 1111 1111
10	2^9	512	1,023	0000 03FF	0011 1111 1111
11	2^{10}	1,024	2,047	0000 07FF	0111 1111 1111
12	2^{11}	2,048	4,095	0000 0FFF	1111 1111 1111
13	2^{12}	4,096	8,191	0000 1FFF	etc.
14	2^{13}	8,192	16,383	0000 3FFF	etc.
15	2^{14}	16,384	32,767	0000 7FFF	etc.
16	2^{15}	32,768	65,535	0000 FFFF	-

Find the nearest
power of 2

1000 needed



Powers of 2 Table					
Bit Line #	Powers of 2 Expo- nent	Binary Bit Weight in Decimal	Highest Number Count (Memory Address)		
			Decimal	Hexadecimal	Binary
1	2^0	1	1	0001	0000 0001
2	2^1	2	3	0003	0000 0011
3	2^2	4	7	0007	0000 0111
4	2^3	8	15	000F	0000 1111
...	...	...	...	...	...
9	2^8	256	511	0000 01FF	0001 1111 1111
10	2^9	512	1,023	0000 03FF	0011 1111 1111
11	2^{10}	1,024	2,047	0000 07FF	0111 1111 1111
12	2^{11}	2,048	4,095	0000 0FFF	1111 1111 1111
13	2^{12}	4,096	8,191	0000 1FFF	etc.
14	2^{13}	8,192	16,383	0000 3FFF	etc.
15	2^{14}	16,384	32,767	0000 7FFF	etc.
16	2^{15}	32,768	65,535	0000 FFFF	- 10

Change IP Address octets to Binary

198.16.32.0



198.16.00100000.00000000

1000 needed

Verify Reserved bits in IP Address

198.16.32.0



198.16.00100000.00000000

1000 needed

2^{10} selected (10 bits reserved)

Verify Reserved bits in IP Address

198.16.32.0



198.16.00100000.00000000

1000 needed

2^{10} selected (10 bits reserved)

Prefix: $/(32-10) = /22$

Results

Organization	First possible IP Addr. in Assigned subnet	Last possible IP Addr. in Assigned subnet	Subnet representation (Slash Notation)
Business A	198.16.32.0		198.16.32.0/22
Business B			
Business C			
Business D			

Get last IP Address

?



198.16.00100011.11111111

Last IP address?

Convert all reserved to 1, then to decimal

Get last IP Address

198.16.35.255



198.16.00100011.11111111

Last IP address?

Convert all reserved to 1, then to decimal

Results

Organization	First possible IP Addr. in Assigned subnet	Last possible IP Addr. in Assigned subnet	Subnet representation (Slash Notation)
Business A	198.16.32.0	198.16.35.255	198.16.32.0/22
Business B			
Business C			
Business D			

Business B

Last IP Address

198.16.35.255

Can't use this one (used by Business A).

Get starting IP Address

198.16.35.255

Can't use this one (used by Business A).
Can we use 198.16.36.0 ?

Find the nearest
power of 2

3000 needed

Powers of 2 Table

Bit Line #	Powers of 2 Expo- nent	Binary Bit Weight in Decimal	Highest Number Count (Memory Address)		
			Decimal	Hexadecimal	Binary
1	2^0	1	1	0001	0000 0001
2	2^1	2	3	0003	0000 0011
3	2^2	4	7	0007	0000 0111
4	2^3	8	15	000F	0000 1111
...	...	...	...	...	...
9	2^8	256	511	0000 01FF	0001 1111 1111
10	2^9	512	1,023	0000 03FF	0011 1111 1111
11	2^{10}	1,024	2,047	0000 07FF	0111 1111 1111
12	2^{11}	2,048	4,095	0000 0FFF	1111 1111 1111
13	2^{12}	4,096	8,191	0000 1FFF	etc.
14	2^{13}	8,192	16,383	0000 3FFF	etc.
15	2^{14}	16,384	32,767	0000 7FFF	etc.
16	2^{15}	32,768	65,535	0000 FFFF	- ²¹

Find the nearest
power of 2

3000 needed



Powers of 2 Table

Bit Line #	Powers of 2 Expo- nent	Binary Bit Weight in Decimal	Highest Number Count (Memory Address)		
			Decimal	Hexadecimal	Binary
1	2^0	1	1	0001	0000 0001
2	2^1	2	3	0003	0000 0011
3	2^2	4	7	0007	0000 0111
4	2^3	8	15	000F	0000 1111
...	...	...	...	...	...
9	2^8	256	511	0000 01FF	0001 1111 1111
10	2^9	512	1,023	0000 03FF	0011 1111 1111
11	2^{10}	1,024	2,047	0000 07FF	0111 1111 1111
12	2^{11}	2,048	4,095	0000 0FFF	1111 1111 1111
13	2^{12}	4,096	8,191	0000 1FFF	etc.
14	2^{13}	8,192	16,383	0000 3FFF	etc.
15	2^{14}	16,384	32,767	0000 7FFF	etc.
16	2^{15}	32,768	65,535	0000 FFFF	- 22

Get starting IP Address

198.16.35.255

Can't use this one (used by Business A).

Can we use 198.16.36.0 ?

We need 3000

- 2^{12} (12 bits reserved)
- 198.16.36.0 ► 198.16.00100100.00000000

Get starting IP Address

198.16.36.0

Shift bit in third octet until reserved bits are all zeroes:

- 198.16.00100**100.00000000**
- 198.16.001**10000.00000000**

Convert binary back to decimal

- = **198.16.48.0**

Change IP Address octets to Binary

198.16.48.0

Change IP Address octets to Binary

198.16.48.0



198.16.00110000.00000000

3000 needed

Verify Reserved bits in IP Address

198.16.48.0



198.16.00110000.00000000

3000 needed

2^{12} selected (12 bits reserved)

Verify Reserved bits in IP Address

198.16.48.0



198.16.00110000.00000000

3000 needed

2^{12} selected (12 bits reserved)

Prefix: $/(32-12) = /20$

Get last IP Address

?



198.16.00111111.11111111

Last IP address?

Convert all reserved to 1, then to decimal

Get last IP Address

198.16.63.255



198.16.00111111.11111111

Last IP address?

Convert all reserved to 1, then to decimal

Results

Organization	First possible IP Addr. in Assigned subnet	Last possible IP Addr. in Assigned subnet	Subnet representation (Slash Notation)
Business A	198.16.32.0	198.16.35.255	198.16.32.0/22
Business B	198.16.48.0	198.16.63.255	198.16.48.0/20
Business C			
Business D			

Business C

Last IP Address

198.16.63.255

Can't use this one (used by Business B).

Can we use 198.16.64.0 ?

Find the nearest power of 2

2000 needed

Powers of 2 Table

Bit Line #	Powers of 2 Exponent	Binary Bit Weight in Decimal	Highest Number Count (Memory Address)		
			Decimal	Hexadecimal	Binary
1	2^0	1	1	0001	0000 0001
2	2^1	2	3	0003	0000 0011
3	2^2	4	7	0007	0000 0111
4	2^3	8	15	000F	0000 1111
...	...	...	...	...	...
9	2^8	256	511	0000 01FF	0001 1111 1111
10	2^9	512	1,023	0000 03FF	0011 1111 1111
11	2^{10}	1,024	2,047	0000 07FF	0111 1111 1111
12	2^{11}	2,048	4,095	0000 0FFF	1111 1111 1111
13	2^{12}	4,096	8,191	0000 1FFF	etc.
14	2^{13}	8,192	16,383	0000 3FFF	etc.
15	2^{14}	16,384	32,767	0000 7FFF	etc.
16	2^{15}	32,768	65,535	0000 FFFF	- 34

Find the nearest
power of 2

2000 needed



Powers of 2 Table					
Bit Line #	Powers of 2 Expo- nent	Binary Bit Weight in Decimal	Highest Number Count (Memory Address)		
			Decimal	Hexadecimal	Binary
1	2^0	1	1	0001	0000 0001
2	2^1	2	3	0003	0000 0011
3	2^2	4	7	0007	0000 0111
4	2^3	8	15	000F	0000 1111
...	...	...	...	...	...
9	2^8	256	511	0000 01FF	0001 1111 1111
10	2^9	512	1,023	0000 03FF	0011 1111 1111
11	2^{10}	1,024	2,047	0000 07FF	0111 1111 1111
12	2^{11}	2,048	4,095	0000 0FFF	1111 1111 1111
13	2^{12}	4,096	8,191	0000 1FFF	etc.
14	2^{13}	8,192	16,383	0000 3FFF	etc.
15	2^{14}	16,384	32,767	0000 7FFF	etc.
16	2^{15}	32,768	65,535	0000 FFFF	- 35

Get starting IP Address

198.16.63.255

Can we use 198.16.64.0 ?

We know that we need 2000:

- 2^{11} (11 bits reserved)
- 198.16.64.0 ► 198.16.01000000.00000000

Get starting IP Address

198.16.63.255

198.16.64.0 has all 11 bits available, so we can use it.

Starting IP Address

198.16.64.0

Change IP Address octets to Binary

198.16.64.0



198.16.01000000.00000000

2000 needed

Change IP Address octets to Binary

198.16.64.0



198.16.01000000.00000000

2000 needed

2^{11} selected (11 bits reserved)

Change IP Address octets to Binary

198.16.64.0



198.16.01000000.00000000

2000 needed

2^{11} selected (11 bits reserved)

Prefix: $/ (32 - 11) = /21$

Get last IP Address

?



198.16.01000111.11111111

Last IP address?

Convert all reserved to 1, then to decimal

Get last IP Address

198.16.71.255



198.16.01000111.11111111

Last IP address?

Convert all reserved to 1, then to decimal

Results

Organization	First possible IP Addr. in Assigned subnet	Last possible IP Addr. in Assigned subnet	Subnet representation (Slash Notation)
Business A	198.16.32.0	198.16.35.255	198.16.32.0/22
Business B	198.16.48.0	198.16.63.255	198.16.48.0/20
Business C	198.16.64.0	198.16.71.255	198.16.64.0/21
Business D			

Business D

Last IP Address

198.16.71.255

Can we use 198.16.72.0 ?

Last IP Address

198.16.71.255

Can we use 198.16.72.0 ?

We need 4500

Find the nearest
power of 2

4500 needed

Powers of 2 Table

Bit Line #	Powers of 2 Expo- nent	Binary Bit Weight in Decimal	Highest Number Count (Memory Address)		
			Decimal	Hexadecimal	Binary
1	2^0	1	1	0001	0000 0001
2	2^1	2	3	0003	0000 0011
3	2^2	4	7	0007	0000 0111
4	2^3	8	15	000F	0000 1111
...	...	...	...	...	...
9	2^8	256	511	0000 01FF	0001 1111 1111
10	2^9	512	1,023	0000 03FF	0011 1111 1111
11	2^{10}	1,024	2,047	0000 07FF	0111 1111 1111
12	2^{11}	2,048	4,095	0000 0FFF	1111 1111 1111
13	2^{12}	4,096	8,191	0000 1FFF	etc.
14	2^{13}	8,192	16,383	0000 3FFF	etc.
15	2^{14}	16,384	32,767	0000 7FFF	etc.
16	2^{15}	32,768	65,535	0000 FFFF	- 48

Find the nearest
power of 2

4500 needed



Powers of 2 Table

Bit Line #	Powers of 2 Expo- nent	Binary Bit Weight in Decimal	Highest Number Count (Memory Address)		
			Decimal	Hexadecimal	Binary
1	2^0	1	1	0001	0000 0001
2	2^1	2	3	0003	0000 0011
3	2^2	4	7	0007	0000 0111
4	2^3	8	15	000F	0000 1111
...	...	...	...	...	...
9	2^8	256	511	0000 01FF	0001 1111 1111
10	2^9	512	1,023	0000 03FF	0011 1111 1111
11	2^{10}	1,024	2,047	0000 07FF	0111 1111 1111
12	2^{11}	2,048	4,095	0000 0FFF	1111 1111 1111
13	2^{12}	4,096	8,191	0000 1FFF	etc.
14	2^{13}	8,192	16,383	0000 3FFF	etc.
15	2^{14}	16,384	32,767	0000 7FFF	etc.
16	2^{15}	32,768	65,535	0000 FFFF	- 49

Get starting IP Address

198.16.71.255

Can we use 198.16.72.0 ?

We need 4500

- 2^{13} (13 bits reserved)
- 198.16.72.0 ► 198.16.01001000.00000000

Get starting IP Address

198.16.71.255

Shift 1 bit in third octet until reserved bits are all zeroes:

- 198.16.01001000.00000000
- 198.16.01100000.00000000
- = **198.16.96.0**

Starting IP Address

198.16.96.0

Change IP Address octets to Binary

198.16.96.0



198.16.01100000.00000000

4500 needed

2^{13} selected (13 bits reserved)

Change IP Address octets to Binary

198.16.96.0



198.16.01100000.00000000

4500 needed

2^{13} selected (13 bits reserved)

Prefix: $/(32-13) = /19$

Get last IP Address

?



198.16.01111111.11111111

Last IP address?

Convert all reserved to 1, then to decimal

Get last IP Address

198.16.127.255



198.16.01111111.11111111

Last IP address?

Convert all reserved to 1, then to decimal

Results

Organization	First possible IP Addr. in Assigned subnet	Last possible IP Addr. in Assigned subnet	Subnet representation (Slash Notation)
Business A	198.16.32.0	198.16.35.255	198.16.32.0/22
Business B	198.16.48.0	198.16.63.255	198.16.48.0/20
Business C	198.16.64.0	198.16.71.255	198.16.64.0/21
Business D	198.16.96.0	198.16.127.255	198.16.96.0/19

Exercise

Exercise

Assume we have a starting address of 190.0.39.128. If company A requests 1500 IP addresses, what would be the starting and ending addresses?

- A. 190.0.40.0 ; 190.0.47.255
- B. 190.0.39.0 ; 190.0.39.255
- C. 190.0.39.255 ; 190.0.47.255
- D. 190.0.40.0 ; 190.0.40.255
- E. 190.0.39.0 ; 190.0.40.255

Given IP Address

190.0.39.128

Can we use it?

Last IP Address

190.0.39.128

Can we use it?

We need 1500 addresses

Find the nearest
power of 2

1500 needed

Powers of 2 Table

Bit Line #	Powers of 2 Expo- nent	Binary Bit Weight in Decimal	Highest Number Count (Memory Address)		
			Decimal	Hexadecimal	Binary
1	2^0	1	1	0001	0000 0001
2	2^1	2	3	0003	0000 0011
3	2^2	4	7	0007	0000 0111
4	2^3	8	15	000F	0000 1111
...	...	...	...	...	...
9	2^8	256	511	0000 01FF	0001 1111 1111
10	2^9	512	1,023	0000 03FF	0011 1111 1111
11	2^{10}	1,024	2,047	0000 07FF	0111 1111 1111
12	2^{11}	2,048	4,095	0000 0FFF	1111 1111 1111
13	2^{12}	4,096	8,191	0000 1FFF	etc.
14	2^{13}	8,192	16,383	0000 3FFF	etc.
15	2^{14}	16,384	32,767	0000 7FFF	etc.
16	2^{15}	32,768	65,535	0000 FFFF	- 62

Find the nearest
power of 2

1500 needed



Powers of 2 Table					
Bit Line #	Powers of 2 Expo- nent	Binary Bit Weight in Decimal	Highest Number Count (Memory Address)		
			Decimal	Hexadecimal	Binary
1	2^0	1	1	0001	0000 0001
2	2^1	2	3	0003	0000 0011
3	2^2	4	7	0007	0000 0111
4	2^3	8	15	000F	0000 1111
...	...	...	...	...	...
9	2^8	256	511	0000 01FF	0001 1111 1111
10	2^9	512	1,023	0000 03FF	0011 1111 1111
11	2^{10}	1,024	2,047	0000 07FF	0111 1111 1111
12	2^{11}	2,048	4,095	0000 0FFF	1111 1111 1111
13	2^{12}	4,096	8,191	0000 1FFF	etc.
14	2^{13}	8,192	16,383	0000 3FFF	etc.
15	2^{14}	16,384	32,767	0000 7FFF	etc.
16	2^{15}	32,768	65,535	0000 FFFF	- 63

Get starting IP Address

190.0.39.128

Can we use it?

We need 1500 addresses

- 2^{11} (11 bits reserved)
- 190.0.39.128 ► 190.0.00100111.10000000

Get starting IP Address

190.0.39.128

Add 1 bit in fourth octet until reserved bits are all zeroes:

- 190.0.00100**111.10000000**
- 190.0.0010**1000.00000000**
- = **190.0.40.0**

Starting IP Address

190.0.40.0

Change IP Address octets to Binary

190.0.40.0



190.0.00101000.00000000

1500 needed

2^{11} selected (11 bits reserved)

Change IP Address octets to Binary

190.0.40.0



190.0.00101000.00000000

1500 needed

2^{11} selected (11 bits reserved)

Prefix: $/(32-11) = /21$

Get last IP Address

?



190.0.00101111.11111111

Last IP address?

Convert all reserved to 1, then to decimal

Get last IP Address

190.0.47.255



190.0.00101111.11111111

Last IP address?

Convert all reserved to 1, then to decimal

Exercise

Assume we have a starting address of 190.0.39.128. If company A requests 1500 IP addresses, what would be the starting and ending addresses?

- A. 190.0.40.0 ; 190.0.47.255
- B. 190.0.39.0 ; 190.0.39.255
- C. 190.0.39.255 ; 190.0.47.255
- D. 190.0.40.0 ; 190.0.40.255
- E. 190.0.39.0 ; 190.0.40.255